WHAT IS THE CHANGE I WANT TO CREATE?

- Barcelona will replace **thousands of plane trees**. It surely has criteria for the new species, but they are **not published**, and no public document says what a choice means for **heat or flooding**.
- For every tree to replace, designers and the city should see which species gives **more summer shade** and **holds back more rain** at that spot.
- Always within the city's rules: **no species above 15%** of the trees, and **bigger trees only on wider sidewalks**.

OUTPUT

- **Shade & Sponge**: an **open-source tool** (Python core, web app, Grasshopper link).
- **In:** tree inventory, LiDAR crowns, terrain, design storm, summer weather file.
- **Out: a species and position** for each replacement, scored on **summer shade** / UTCI and runoff against the **city's own replacement mix**.
- Tested in **Porta** (Nou Barris); run on **33 neighbourhoods**.

SHORT-TERM OUTCOMES

- **Technological:** a reproducible method and open code, checked in **Ladybug and Honeybee**.
- **Environmental (simulated): more summer shade** than the city's real replacement mix in all **33 neighbourhoods**; pedestrian heat stays as with today's planes (**±0.2 °C UTCI**). **Gains:** shade, runoff, diversity, **winter sun**.
- **Knowledge:** species and positions chosen for **heat and runoff together**, not done yet in the literature reviewed.

MEDIUM-TERM OUTCOMES

- **Technological:** designers and **Parcs i Jardins** test it on real replacement campaigns or pilot streets.
- **Replication:** a **transferability test** in two other cities, **Florence and Mataró**, with the same method and their own data.
- **Social:** streets keep their shade and comfort while the planes are replaced, starting where heat and flooding overlap.
- **Economic:** fewer trial-and-error plantings, faster design.

LONG-TERM IMPACT

- **Environmental:** canopy moves toward the city's **30% target for 2037** with **no species above 15%**; less runoff to sewers in flood hotspots.
- **Social: no loss of pedestrian comfort** during the plane replacement, with more canopy where it is lacking.
- **Economic:** avoided drainage and heat-health costs.
- A **replicable method** for cities replacing monocultures.

ASSUMPTIONS

- The city **keeps replacing plane trees** over the next decade (press, 2026: more than half).
- A fast shade proxy and a simple runoff model **rank species correctly**. Partly checked for heat: proxy shade tracks Ladybug UTCI (**r = −0.82**).
- LiDAR crowns and LAI **represent mature trees**; light through a crown depends on its LAI.
- The city **keeps its current rules** (**no species above 15%**, tree size set by sidewalk width): the tool is built on them.
- The city and designers are **willing to use an open tool**.

RISKS

- Optimising two services can concentrate on a few 'best' species: **pest risk** (mitigated by the cap and a genus check).
- **Results over-trusted:** simulated, not measured. The heat difference is small, and its **sign flips** between solid and porous crowns.
- Big crowns vs **winter sun**, façades, roots, maintenance, allergies.
- Benefits arrive only **when trees mature** (decades).
- Used to justify **removing healthy trees**.

POTENTIAL INDICATORS

- % sidewalk in **summer shade** vs city mix (pts)
- **Δ UTCI**, 12–17 h, July day (°C; Ladybug, Honeybee)
- **Δ runoff**, design storm (m³, %)
- **Agreement:** proxy vs Ladybug (r), Python vs Grasshopper (r)
- Neighbourhoods where the tool **beats the city mix** (n / 33)

POTENTIAL INDICATORS

- Streets or projects that **use the tool** (n)
- **Design time** per street, before vs after (h)
- **Measured** air / surface temperature in pilot streets
- Results in **Florence and Mataró** vs **Porta**
- **Designer feedback** (interviews, ease of use)

POTENTIAL INDICATORS

- **Canopy cover** per neighbourhood (LiDAR; **Porta** 20.1% in 2021 vs 30% target)
- Share of the **largest species** (planes ~27% → ≤15%)
- Hours above **strong heat stress** (UTCI > 32 °C)
- **Sewer overflows** and flood reports in hotspot streets
- Cities that **adopt the method**

Shade & Sponge · Theory of Change choosing replacement street trees for shade and runoff (Barcelona, Porta)

Impact level: short-term = demonstrated early (simulation, 33 neighbourhoods); medium and long term = expected.